

- Dual-Master Switching: Uses a low-cost 2:1 Mux to automatically switch control between the PC (Desk Mode) and Arduino (Field Mode).
- USB 3.0 & LTE Support: Integrates the VL817 Hub and A7672S LTE module.
- Protection: Includes full ESD, Over-Current, and Back-Feed protection.

1. System Architecture Diagram

```
graph TD
    %% Inputs
    PC_PORT[External Type-C Port (PC)] -->|USB 3.0 + VBUS| MUX_IN_B
    ARDUINO[Arduino Uno Q] -->|USB 3.0 + VBUS| MUX_IN_A

    %% The Switching Brain (2:1 Mux)
    subgraph "Smart Switching Logic"
        MUX[Mux: PI3USB302]
        LOGIC[SEL Logic: PC VBUS Detect]
        MUX_IN_B --> MUX
        MUX_IN_A --> MUX
        PC_PORT -.->|5V Detect| LOGIC
        LOGIC -->|High = Select PC| MUX
    end

    %% The Hub
    MUX -->|Upstream Data| HUB[Hub IC: VL817]

    %% Outputs (Downstream)
    HUB -->|Port 1 (USB 2.0)| LTE[LTE Module: SIMCom A7672S]
    HUB -->|Port 2 (USB 3.0)| USB_A_1[Type-A Connector 1]
    HUB -->|Port 3 (USB 3.0)| USB_A_2[Type-A Connector 2]
    HUB -->|Port 4 (USB 3.0)| USB_C_OUT[Type-C Downstream]

    %% Power System
    EXT_PWR[12V DC Jack] -->|12V| BUCK[5V 5A Buck Converter]
    BUCK -->|Main 5V Rail| HUB
    BUCK -->|High Current 4V| LTE
```

2. Final Bill of Materials (BOM)

Here is the exact list of components you need to buy. Prices are estimated for the Indian market

A. Core Active Components

Component [\[1\]](#) Part Number Function Approx. Price (INR)

USB 3.0 Hub IC [VL817-Q7](#) (VIA Labs) 4-Port SuperSpeed Controller. Integrates 5V/3.3V LDOs. ¹22080

2:1 Mux Switch [PI3USB302-AZBEX](#) (Diodes Inc) Automatically swaps Master between PC and Arduino. ¹165

LTE Module [SIMCom A7672S](#) 4G LTE Cat 1 connectivity for streaming. ¹15000

Buck Converter [TPS54531](#) (TI) or [XL4015 Module](#) Converts Ext 12V to 5V @ 5A to power the Hub + LTE. (4000y)

B. Protective Components (Mandatory)

ESD Protection [TPD4E05U06](#) (Texas Inst.) 4-Channel TVS Diode Array. Place one at every USB port. ⁴35

VBUS Isolation [SS34](#) (Schottky Diode) 3A/40V Diode. Prevents PC 5V from flowing into the board regulator. ¹15

Power Switch [AP22811](#) (Diodes Inc) Current-limiting load switch for downstream ports (prevents shorts). ¹25

Rev. Polarity [P-MOSFET \(AO3401\)](#) Protects the board if you plug the 12V DC jack backwards. ¹10

3. Detailed Schematic Implementation Guide

Step A: Wiring the 2:1 Mux (PI3USB302)

This is the most critical part of your design.

- Pin SEL (Select): Connect this pin directly to the VBUS (5V) pin of your External PC Type-C Connector. Logic: When PC is unplugged, SEL is floating/Low "" Mux selects Input A (Arduino). Logic: When PC is plugged in, SEL sees 5V "" Mux selects Input B (PC). Note: Add a 100k pull-down resistor to GND on the SEL pin to ensure it stays

Low when nothing is plugged in. [4]

- Data Lines:Route TX/RX and D+/D- from Arduino to Input A.Route TX/RX and D+/D- from External PC Port to Input B.Route Output directly to the Upstream pins of the VL817 Hub.

Step B: Protection Wiring

1. VBUS Back-Feed Protection:Place a SS34 Schottky Diode in series with the VBUS line coming from the PC Port.Direction: Anode to PC Port, Cathode to Mux detection logic.Why? This allows the Mux to sense the PC is present (5V), but prevents your board's internal 5V regulator from pushing current out into your PC's motherboard.
2. ESD Arrays (TPD4E05U06):Place these chips physically as close as possible to the metal shell of every USB connector (Input and Outputs).Route the D+, D-, TX, and RX lines through the ESD pads before they go to the Hub/Mux.

Step C: Powering the LTE Module

- Do not power the A7672S from the main 5V rail.
- Use a voltage divider or dedicated LDO on your buck converter to create a 3.8V or 4.0V rail specifically for the VBAT pins of the LTE module.
- Add a 470 μ F Tantalum Capacitor right next to the LTE module's power pins to handle the 2A transmission spikes.

4. Layout & Routing Checklist (KiCad)

- Impedance: Set your generic USB 3.0 trace width for 90-ohm differential impedance. In a standard 4-layer stackup (JLC2313), this is typically:Trace Width: 0.18 mm (7 mil)Trace Spacing: 0.15 mm (6 mil)
- AC Coupling: Place 100nF capacitors on the TX lines of every USB 3.0 link (between Hub and Connector).
- Grounding: Use a solid, unbroken Ground plane on Layer 2 (directly under the Top Signal layer). Do not cut the ground plane under high-speed USB traces.